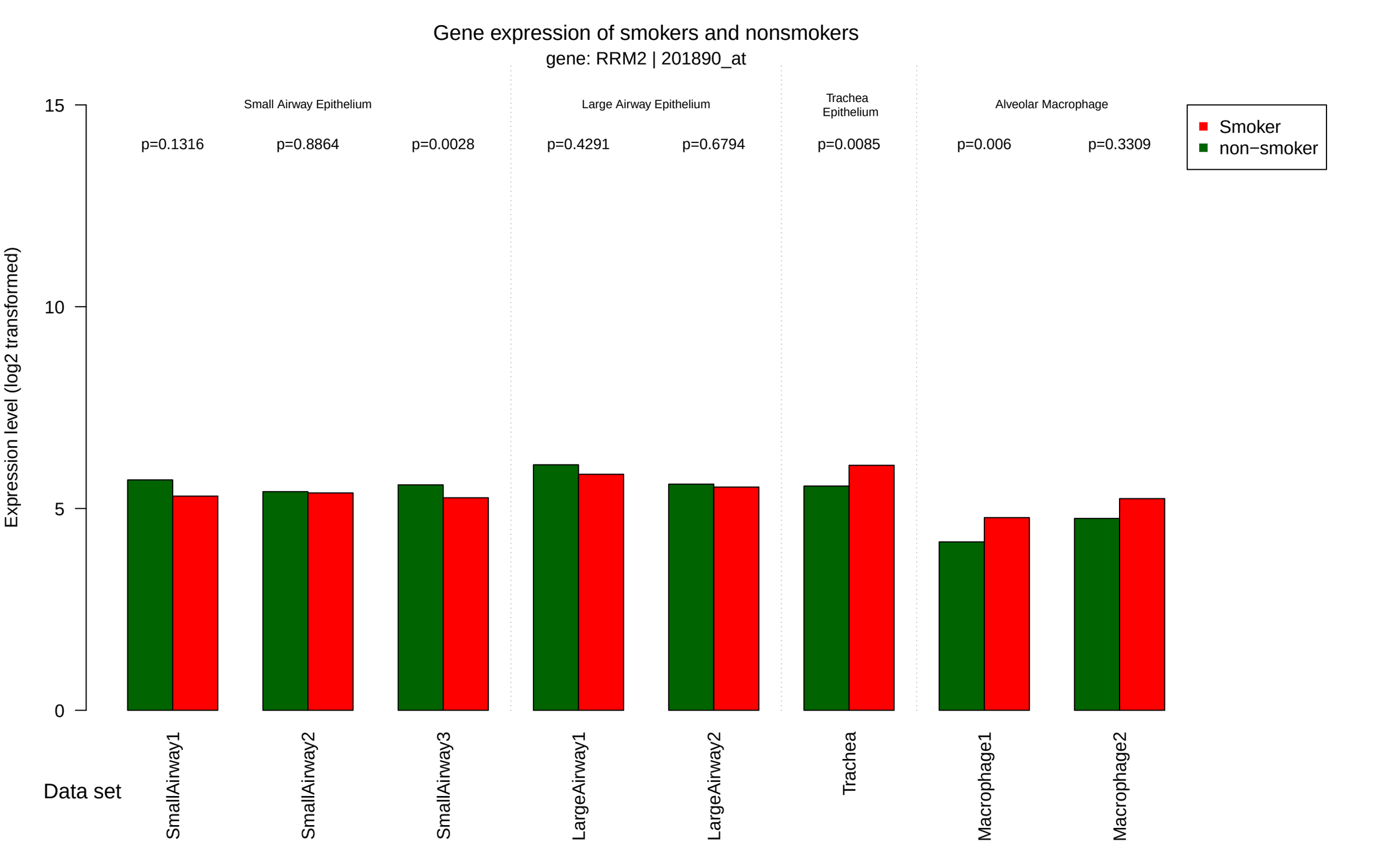
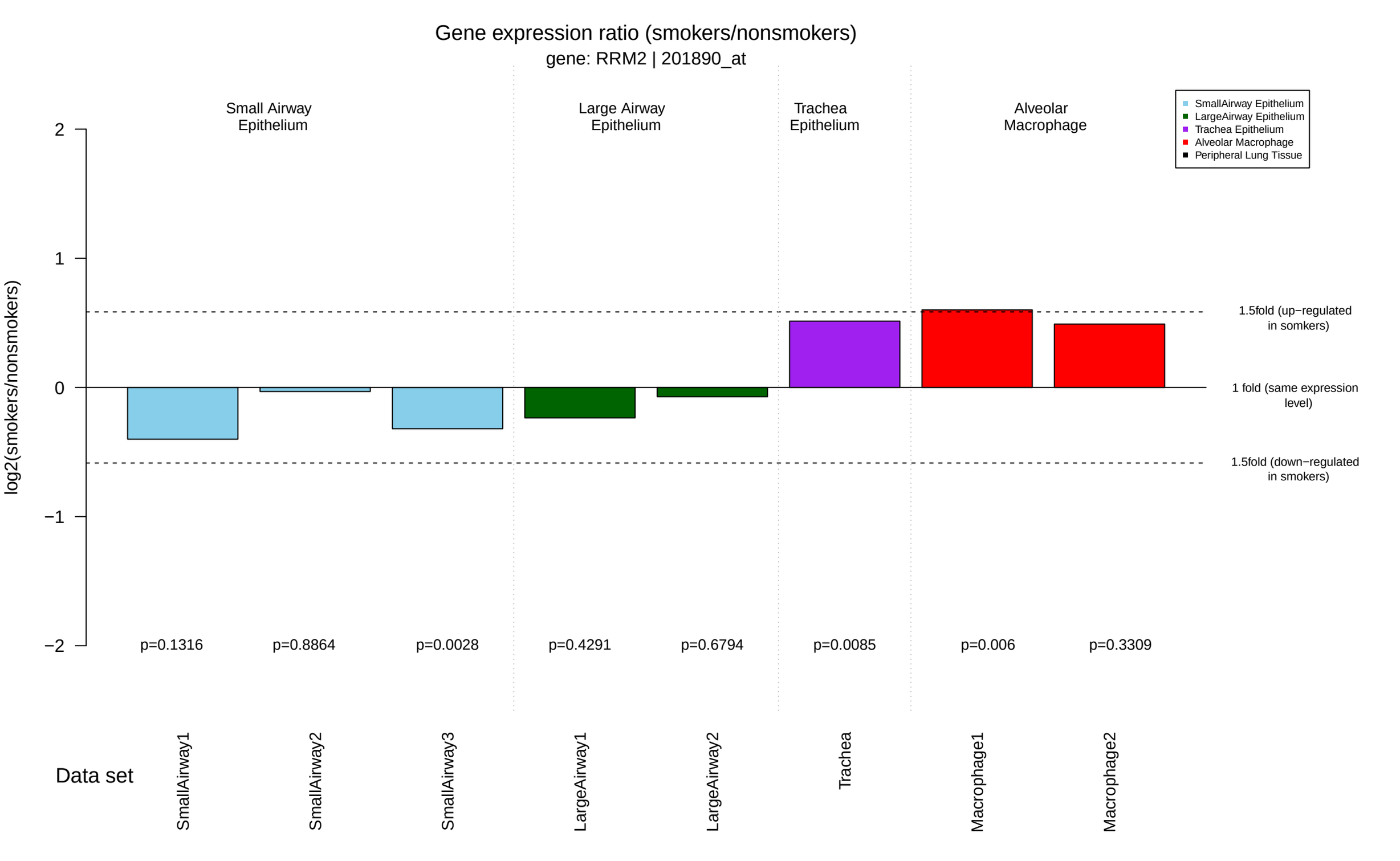
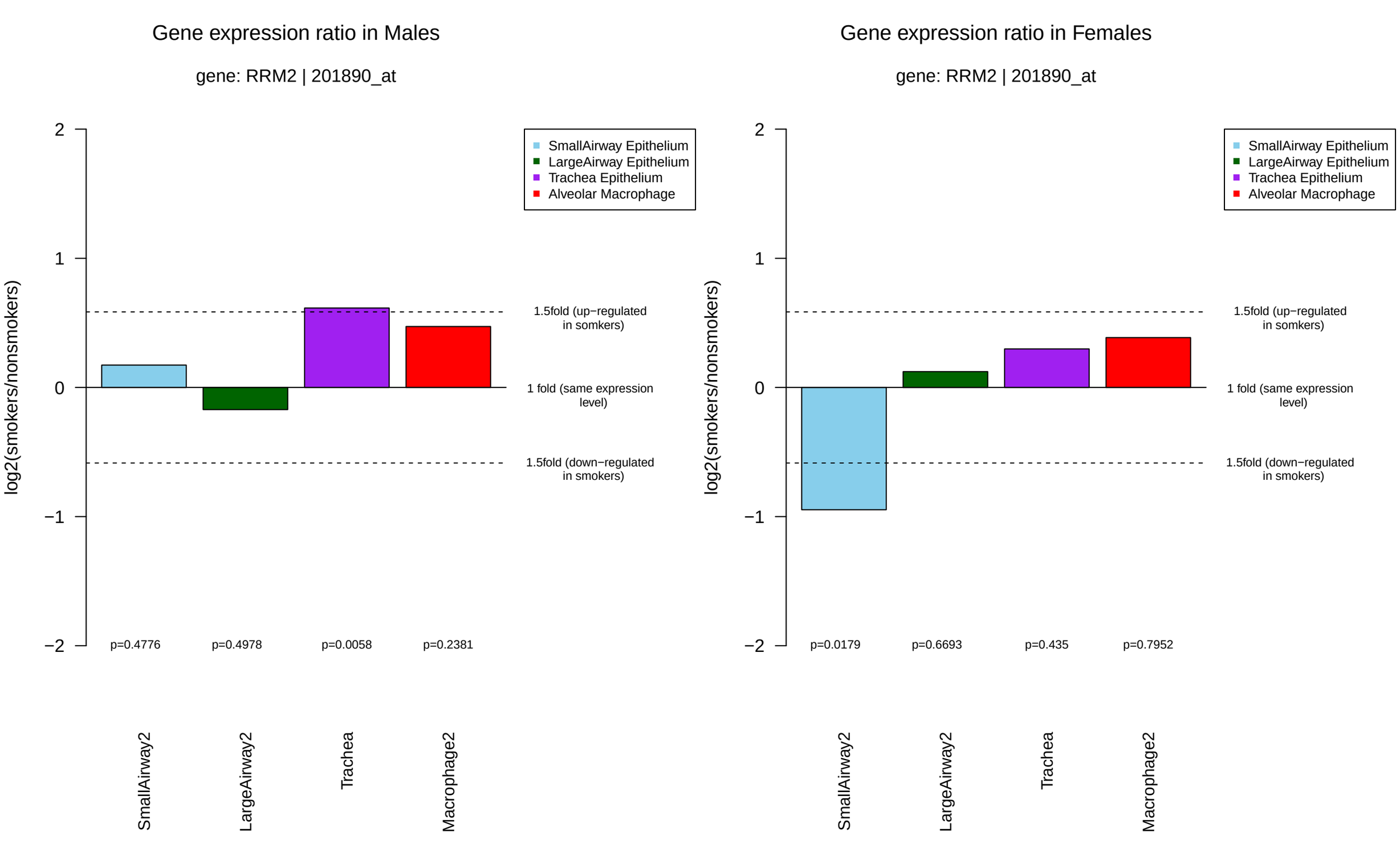
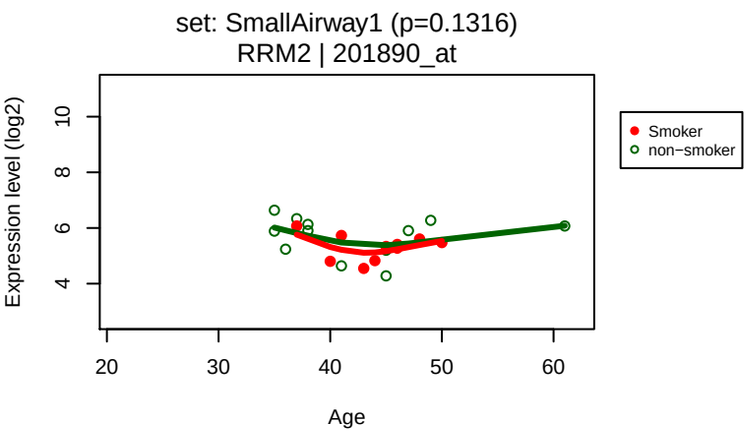
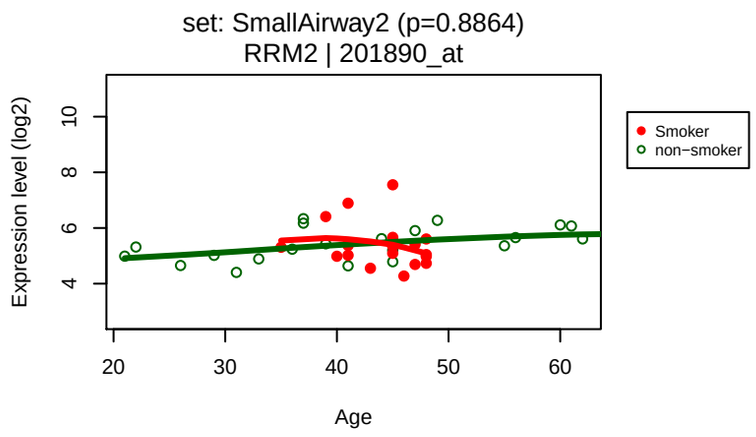
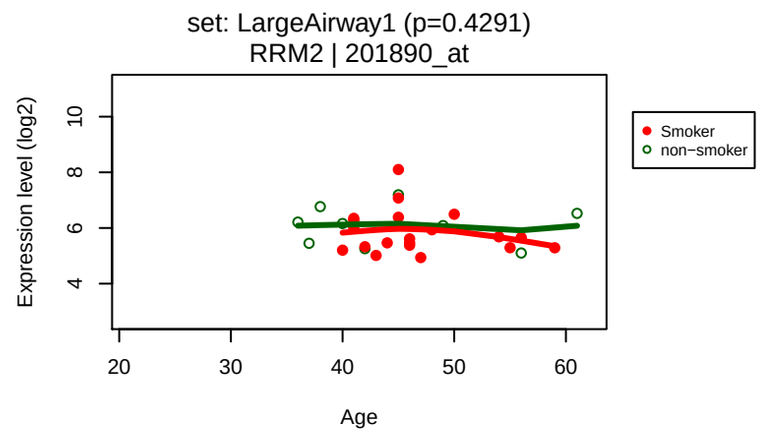

Gene expression of smokers and nonsmokers
gene: RRM2 | 201890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1316
p=0.8864
p=0.0028
p=0.4291
p=0.6794
p=0.0085
p=0.006
p=0.3309
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

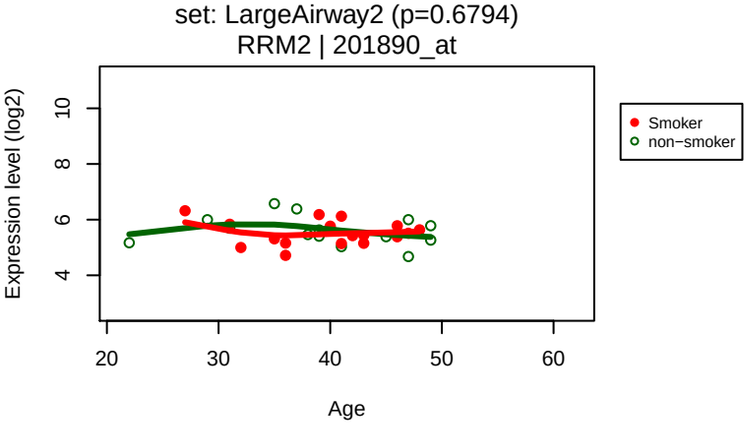

Gene expression ratio (smokers/nonsmokers)
gene: RRM2 | 201890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1316
p=0.8864
p=0.0028
p=0.4291
p=0.6794
p=0.0085
p=0.006
p=0.3309
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

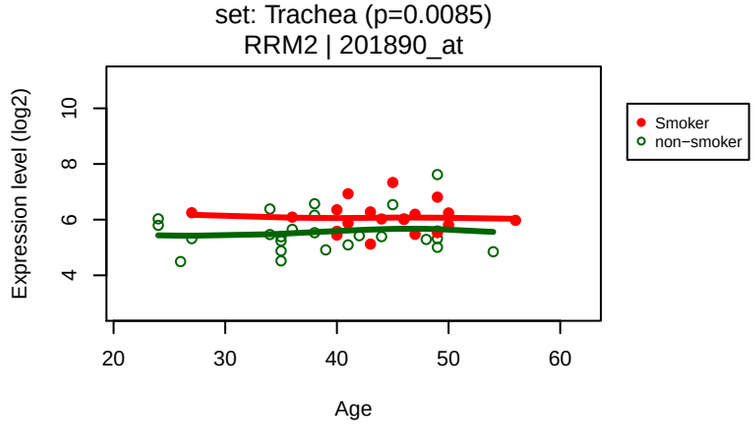

Gene expression ratio in Males
gene: RRM2 | 201890_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4776
p=0.4978
p=0.0058
p=0.2381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRM2 | 201890_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0179
p=0.6693
p=0.435
p=0.7952
SmallAirway2
LargeAirway2
Trachea
Macrophage2

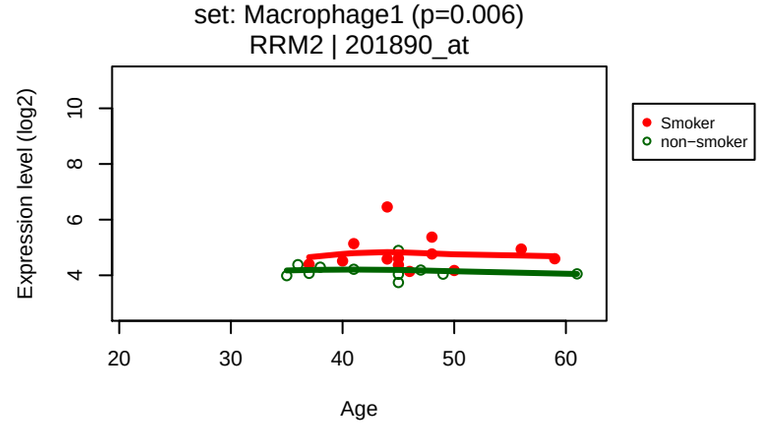

set: SmallAirway1 (p=0.1316)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker

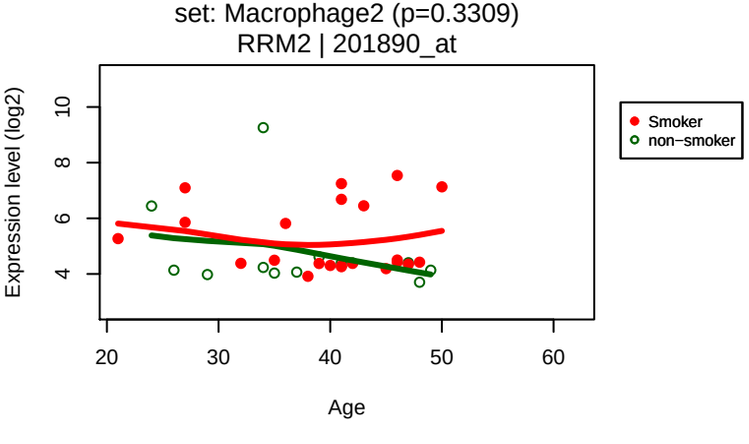

set: SmallAirway2 (p=0.8864)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker

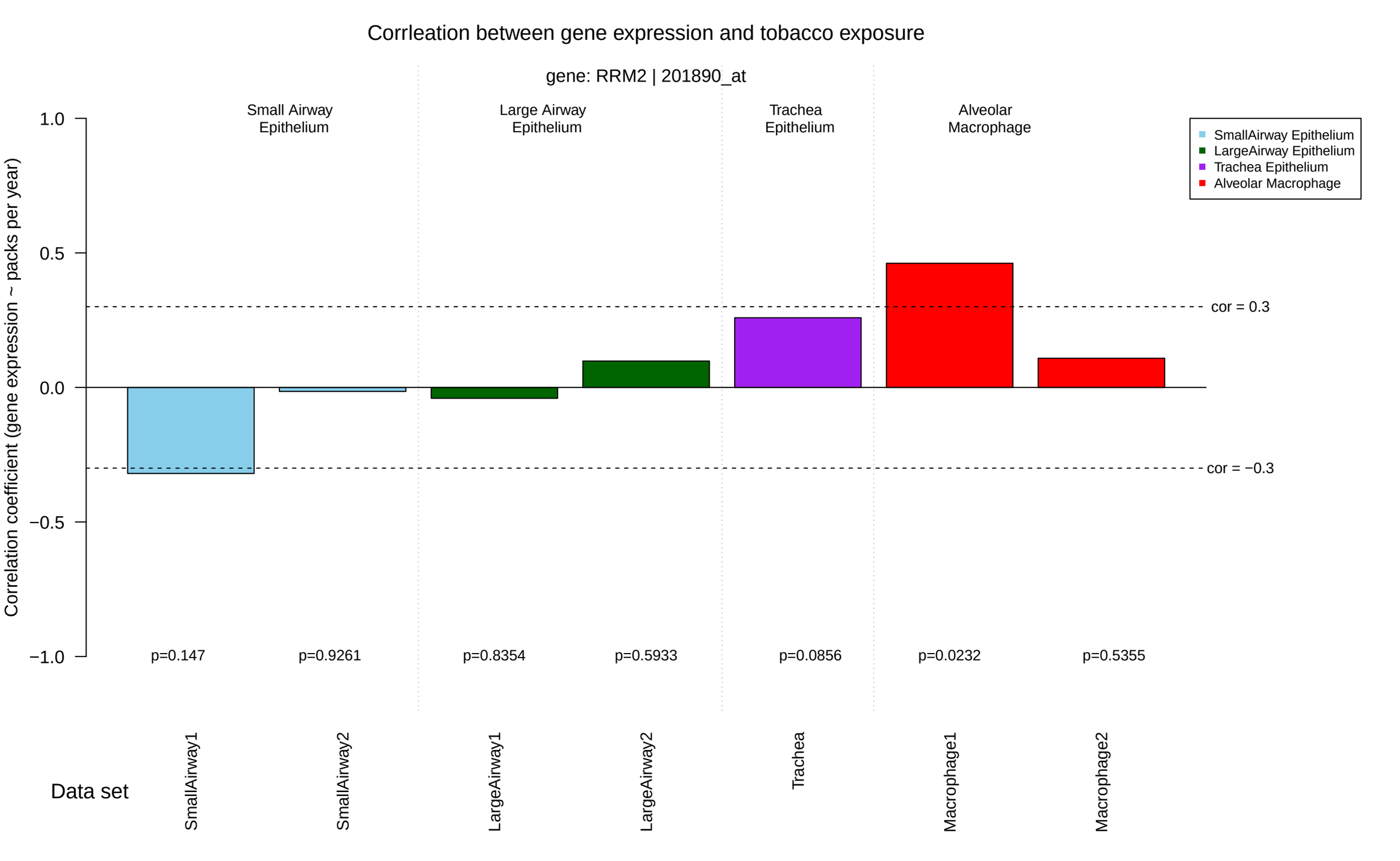

set: LargeAirway1 (p=0.4291)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6794)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0085)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.006)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3309)
RRM2 | 201890_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRM2 | 201890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.147
p=0.9261
p=0.8354
p=0.5933
p=0.0856
p=0.0232
p=0.5355
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2